

Rahul Ashok Jeyalakshmi

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SUMMARY

Results-driven Industrial and Mechanical Engineering professional with hands-on experience in manufacturing systems, process optimization, and equipment maintenance. Proven track record in reducing downtime, improving facility layouts, and implementing lean and Six Sigma methodologies to enhance operational efficiency. Skilled in applying data-driven approaches to optimize production workflows and drive continuous improvement. Passionate about leveraging technical expertise and problem-solving skills to support high-quality, sustainable manufacturing operations.

SKILLS

Root Cause Analysis (RCA), FMEA, Lean Manufacturing, 5S, Six Sigma, SolidWorks, AutoCAD Mechanical, Fusion 360, Python, SQL, C, C++, Power BI, Tableau, MS Excel, Minitab, Process Optimization, Facility Layout Design, Time Studies, Technical Documentation, CAPEX Project Support, Microsoft Office Suite.

PROFESSIONAL EXPERIENCES

University at Buffalo

Feb 2025 - Present

Research Intern

Buffalo, New York, United States

- Identified that 80% of urban freight originates or ends in metropolitan areas, emphasizing the need for emission-focused strategies.
- Assessed logistics influencer's impact on vehicle procurement policies, pushing the adoption of low-emission vehicle fleets.
- Investigated truck companies shift to alternative fuels, linking it to long-term cost savings and lower environmental penalties.
- Compared freight decarbonization trends across Colombia, Brazil, and Turkey with a focus on stakeholder influence.
- Evaluated barriers and enablers for electric vehicle (EV) adoption in Turkey's urban freight sector, emphasizing stakeholder collaboration for decarbonization.

MABS Associates

Jun 2022 - Mar 2023

Industrial Service Engineer

Trichy, Tamil Nadu, India

- Conducted on-site installations, maintenance, and repairs of industrial pumps and motors, reducing time by 10% through standard operating procedures for the production line.
- Collaborated with a team of three members to optimize the maintenance process, cutting repair time by 30% and saving up to \$5,000.
- Implemented preventive maintenance schedules, increasing the lifespan of industrial equipment and lowering operating costs, resulting in improved equipment reliability and reduced unexpected breakdowns.
- Achieved 25% reduction in equipment downtime by utilizing root cause analysis to optimize pump maintenance and enhance overall operational productivity.

PROJECTS

Analyzing the Environmental and Social Impact of Sustainable Practices

Sep 2023 - Dec 2023

- Developed data-driven models using linear regression analysis to forecast environmental benefits and cost savings from implementing green technologies in E-commerce operations.
- Analyzed emerging green supply chain technologies, resulting in a 20% projected reduction in carbon emissions for proposed E-commerce strategies.
- Evaluated carbon footprint reduction formulas and logistics strategies, including CO2 emissions comparisons across air, road, and sea transport.

Optimization of Retail Store Layout Using Market Basket Analysis and A-Score Efficiency

Sep 2023 - Dec 2023

- Conducted a Facility Layout Optimization for Saideep Superstore, enhancing product visibility and operational efficiency.
- Applied Market Basket Analysis to uncover hidden product associations and improve customer flow.
- Utilized A-Score efficiency method to evaluate layout effectiveness, improving layout efficiency from 61.5% to 81.4%.

EDUCATION

University at Buffalo , School of Engineering and Applied Sciences

Jan 2025

Master of Science, Industrial Engineering

GPA: 3.46/4.0

Coursework: Sustainable Manufacturing, Facilities Design, Work Physiology, Project Management, Quality Assurance, E-Business, and Supply Chain Management, Computer-Integrated Manufacturing.

KPR Institute of Engineering and Technology

May 2022

Bachelor of Engineering, Mechanical Engineering

GPA: 3.38/4.0

Coursework: Design of Machine Elements, Lean Manufacturing, Thermodynamics, Manufacturing Engineering, and Strength of Materials.

CERTIFICATIONS

Digital Manufacturing and Design Technology, Lean Six Sigma Green Belt, Lean Expert Certification, Design of Experiments, Supply Chain Sustainability, Certified Value Stream Mapping.

PUBLICATION

[Performance study on solar still using nano disbanded phase change material \(NDPCM\)](#)

ACHIEVEMENTS

Runner-up - Engineering Drawing Competition, awarded by KPR Institute of Engineering and Technology (**Jan 2019**).

Third Place – Moto Quiz Competition, awarded by KPR Institute of Engineering and Technology (**Feb 2020**).

Runner-up – Research Paper Presentation, awarded by Dr. N.G.P. Institute of Technology (**Mar 2022**).